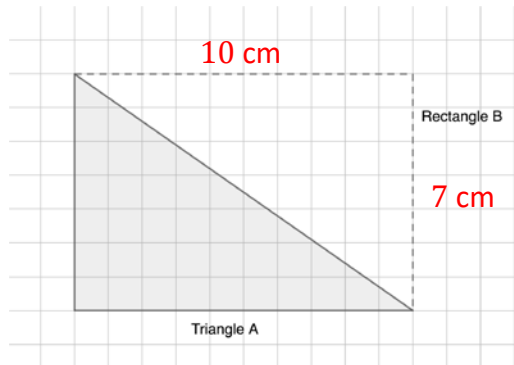


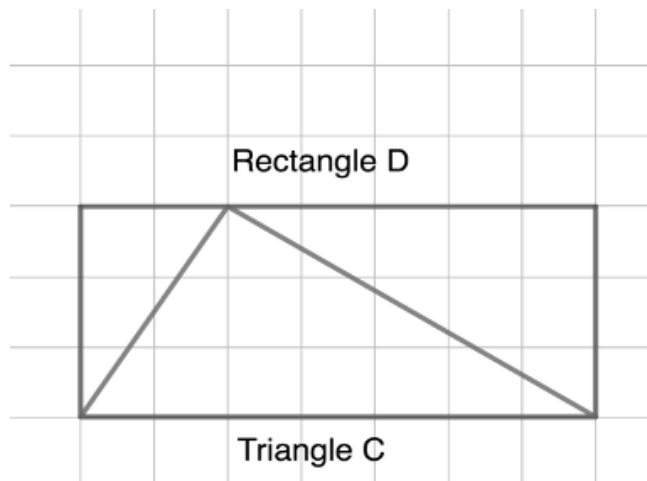
Figure 1 shows **right triangle** *A*, within Rectangle *B*.



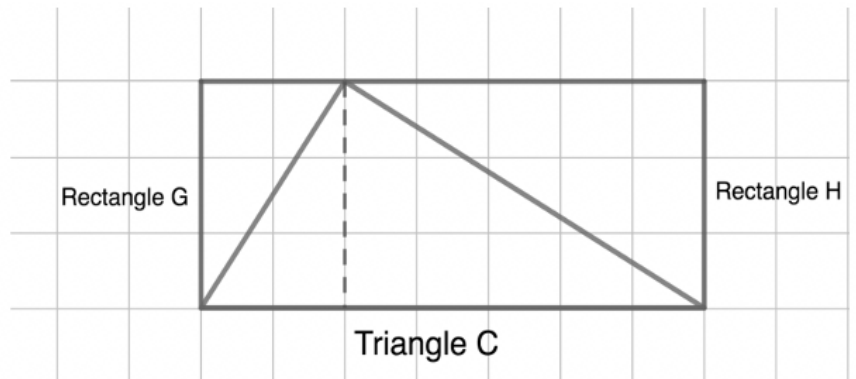
Use the figure to answer questions 1-3. Each square of the grid is 1 cm by 1 cm.

1. It takes 2 triangles to form Rectangle *B*.
2. The area of Rectangle *B* is 70 cm^2 .
7(10)
3. The area of Triangle *A* is 35 cm^2 .
 $\frac{1}{2}(70)$

The figure shows an **acute triangle**, Triangle *C*. Rectangle *D* is drawn around Triangle *C*.



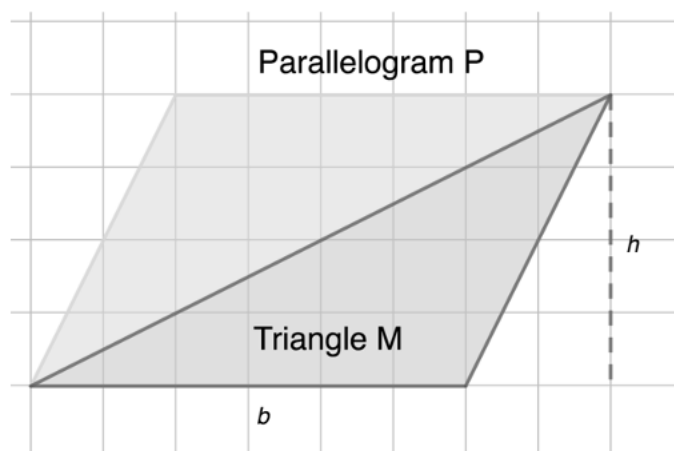
The figure can be broken into smaller rectangles, Rectangle G and Rectangle H , and two right triangles.



Use the figures to answer questions 4-7. Each square of the grid is 1 cm by 1 cm.

4. The area of Rectangle G is 6 cm^2 .
5. The area of Rectangle H is 15 cm^2 .
6. The area of a right triangle is $\frac{1}{2}$ of the area of a rectangle.
7. The total area of Triangle C is 10.5 cm^2 .
 $\frac{1}{2}(6 + 15) = \frac{1}{2}(21)$

Triangle M is an **obtuse triangle**. Parallelogram P is formed by duplicating Triangle M and rearranging it.



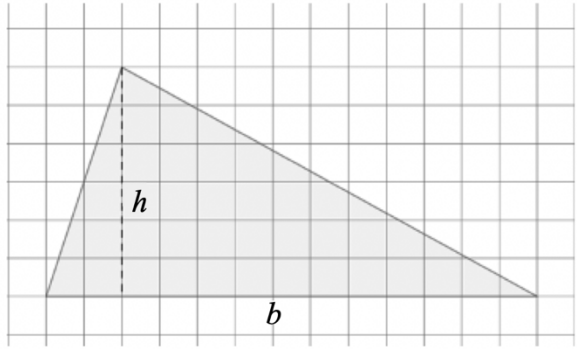
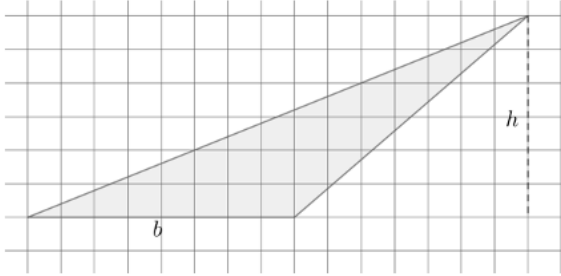
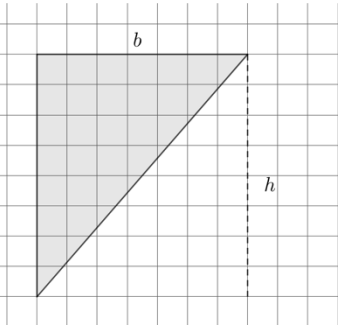
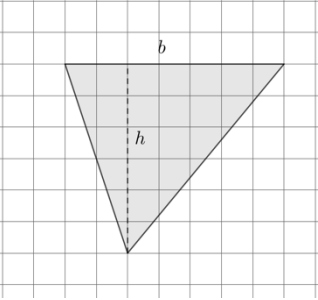
Use the figures to answer questions 8-10. Each square of the grid is 1 cm by 1 cm.

8. The area of Parallelogram P is 24 cm^2 .
9. Triangle M represents $\frac{1}{2}$ of the figure of Parallelogram P .
10. The area of Triangle M is 12 cm^2 .

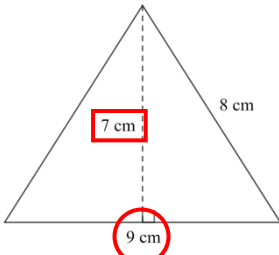
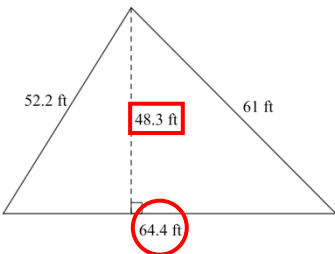
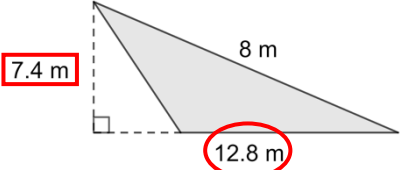
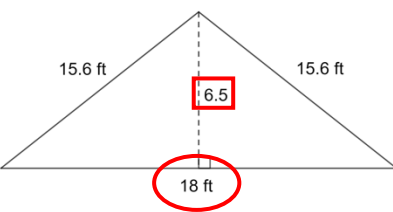
Grade 6 Accelerated
Unit 7
Lesson 2 Homework

Name Answer Key
Date _____
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Four triangles are shown on the grids. For each triangle, a base and its corresponding height are labeled. The side length of each square is 1 centimeter (cm). Find the area of each triangle. Show your work and include units.

1.		$\frac{1}{2} \cdot b \cdot h$ $\frac{1}{2} \cdot 6 \cdot 13$ $3 \cdot 13 = 39$ Area = <u>39 cm²</u>
2.		$\frac{1}{2} \cdot 8 \cdot 6$ $4 \cdot 6 = 24$ Area = <u>24 cm²</u>
3.		$\frac{1}{2} \cdot b \cdot h$ $\frac{1}{2} \cdot 7 \cdot 8$ $4 \cdot 7 = 28$ Area = <u>28 cm²</u>
4.		$\frac{1}{2} \cdot b \cdot h$ $\frac{1}{2} \cdot 7 \cdot 6$ $3 \cdot 7 = 21$ Area = <u>21 cm²</u>

For each triangle below, circle the base measurement and put a square around the height measurement. Then, use the base and height to find the area of each triangle. Show your work and include units.

5.		$\frac{1}{2} \cdot 7 \cdot 9$ $3.5 \cdot 9 = 31.5$ Area = <u>31.5 cm²</u>
6.		$\frac{1}{2} \cdot 48.3 \cdot 64.4$ $48.3 \cdot 32.2 = 1555.26$ Area = <u>1,555.26 ft²</u>
7.		$\frac{1}{2} \cdot 7.4 \cdot 12.8$ $7.4 \cdot 6.4 = 47.36$ Area = <u>47.36 m²</u>
8.		$\frac{1}{2} \cdot 18 \cdot 6.5$ $9 \cdot 6.5 = 58.5$ Area = <u>58.5 ft²</u>

Janelle found the area of the triangle shown.

Her work is also shown.

$$\text{Area} = 23.2 \times 16.5 \times 0.5$$

$$\text{Area} = 191.4 \text{ mm}^2$$

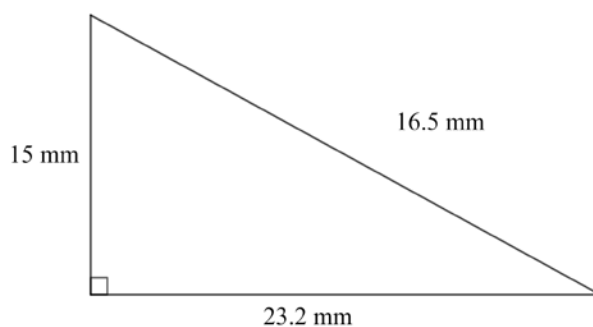
9. Explain why Janelle's answer is incorrect.

Because the height is 15 not 16.5.

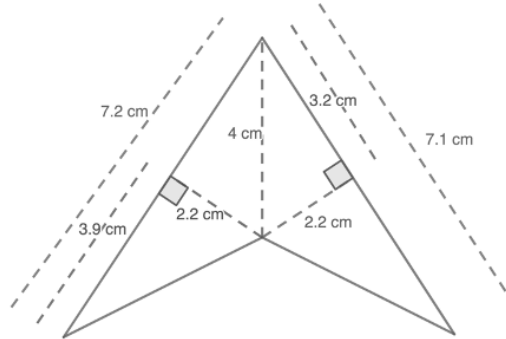
(16.5 is the hypotenuse)

10. Find the correct area of the triangle.

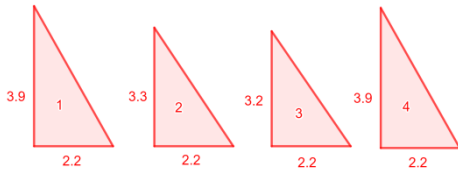
$$\frac{1}{2} \cdot 23.2 \cdot 15 = 174 \text{ mm}^2$$



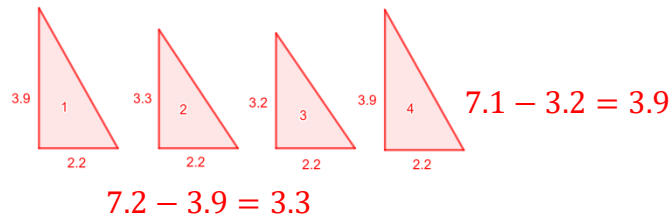
A quadrilateral with measurements in centimeters (cm) is shown for questions 1–3.



1. Decompose the quadrilateral into triangles. Sketch the decomposed shapes here.



2. Using the measurements given in the original image, label the base and height of each triangle.



3. Find the area of the original quadrilateral by finding the sum of the areas of the decomposed triangles. Include units.

$$\triangle 1 = 0.5 \cdot 3.9 \cdot 2.2 = 4.29$$

$$\triangle 2 = 0.5 \cdot 3.3 \cdot 2.2 = 3.63$$

$$\triangle 3 = 0.5 \cdot 2.2 \cdot 3.2 = 3.52$$

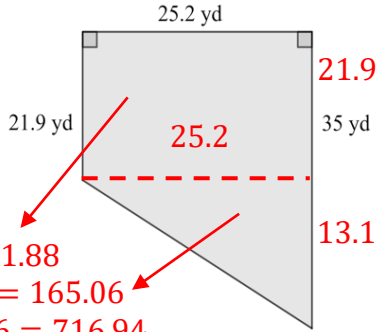
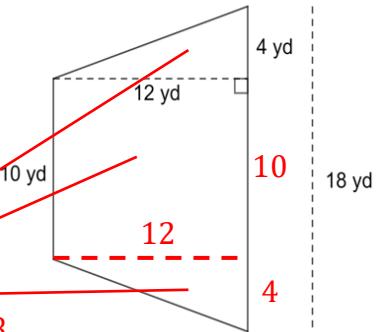
$$\triangle 4 = 0.5 \cdot 2.2 \cdot 3.9 = 4.29$$

$$4.29 + 3.63 + 3.52 + 4.29 = 15.73 \text{ cm}^2$$

Decompose each quadrilateral to find the area. Lengths are shown in inches (in.).

4. $(20 \cdot 15) + (0.5 \cdot 20 \cdot 5) + (0.5 \cdot 20 \cdot 5)$ $300 + 50 + 50$	5. $22 \cdot 20 = 440$ $0.5 \cdot 22 \cdot 14 = 154$ $440 + 154 = 594$	6. $0.5 \cdot 12 \cdot 5.5 = 33$ $0.5 \cdot 4.5 \cdot 12 = 27$ $33 + 27 = 60$
Area = <u>400 in²</u>	Area = <u>594 in²</u>	Area = <u>60 in²</u>

Decompose each quadrilateral to find the area. Lengths are shown in yards (yd).

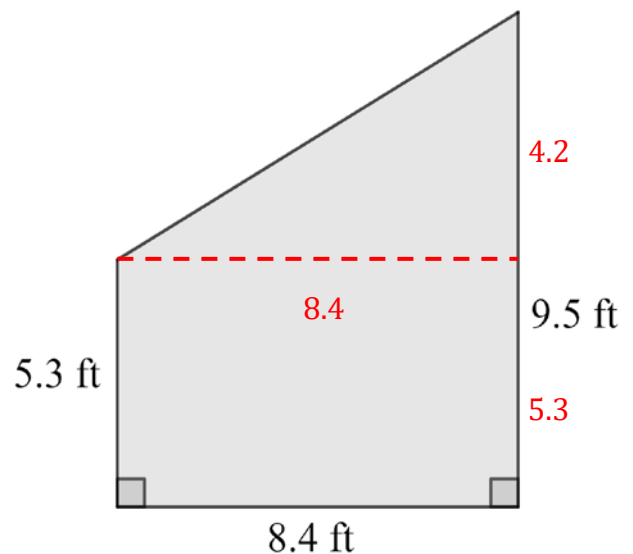
7.  $25.2 \cdot 21.9 = 551.88$ $0.5 \cdot 25.2 \cdot 13.1 = 165.06$ $551.88 + 165.06 = 716.94$ Area = <u>716.94 yd²</u>	8.  $0.5 \cdot 12 \cdot 4 = 24$ $10 \cdot 12 = 120$ $0.5 \cdot 12 \cdot 4 = 24$ $120 + 24 + 24 = 168$ Area = <u>168 yd²</u>
---	---

Sal made a sketch of a wall he needs to paint. The sketch of the wall is shown with measurements in feet for questions 9–10.

9. The home improvement store is offering a sale on their quarts of paint. A quart of paint covers 100 square feet of wall. Will Sal have enough paint if he buys one quart?

$$\begin{aligned} \text{Area Rectangle} &= 5.3 \cdot 8.4 = 44.52 \\ \text{Area Triangle} &= 0.5 \cdot 8.4 \cdot 4.2 = 17.64 \\ \text{Total Area} &= 44.52 + 17.64 = 62.16 \text{ ft}^2 \end{aligned}$$

Sal will have enough because the area of the wall is less than 100 ft².



10. Determine the amount of paint, in square feet, Sal is short or has left over.

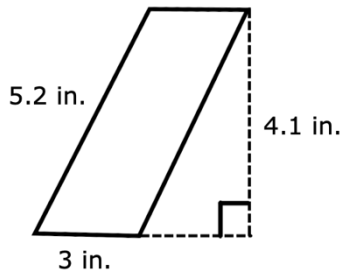
$$\begin{array}{r} 100.00 \\ - 62.16 \\ \hline 37.84 \end{array}$$

He will have 37.84 ft² of paint left over.

Grade 6 Accelerated
Unit 7
Lesson 4 Homework

Name Answer Key
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A parallelogram with dimensions labeled is shown below for questions 1-2.



- Which dimension is not needed to find the area of the parallelogram? Explain.
5.2 is not needed because it represents the side length or slant height but not the height of the parallelogram.
- Fill in the blanks to find the area of the parallelogram.

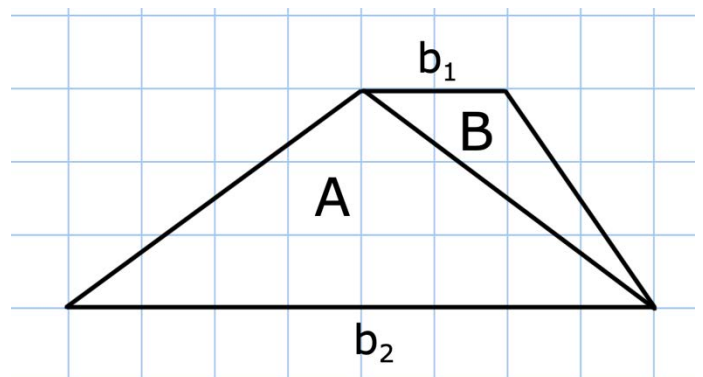
Area of parallelogram = $b \cdot h$

Area = 3 · 4.1

Area = 12.3 in.²

A trapezoid can be decomposed into two triangles with the same height. Use the diagram below for questions 3-6.

- The base of triangle A, b_2 , is 8 units and its corresponding height is 3 units. Therefore, the area of triangle A is 12.
 $\left(\frac{1}{2} \cdot 3 \cdot 8\right)$
- The base of triangle B, b_1 , is 2 units and its corresponding height is 3 units. Therefore, the area of triangle B is 3.
 $\left(\frac{1}{2} \cdot 3 \cdot 2\right)$



- The area of the trapezoid is 12 + 3 = 15 units².

6. Use the formula for area of a trapezoid to verify your solution is correct.

$$\text{Area} = \frac{1}{2}h(b_1 + b_2).$$

$$\text{Area} = \frac{1}{2} \underline{3} (\underline{2} + \underline{8})$$

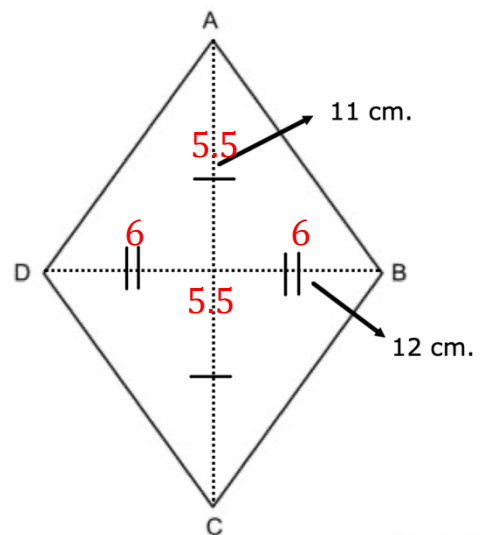
$$\text{Area} = \underline{15} \text{ units}^2$$

Rhombus $ABCD$ is shown. Diagonal BD is 12 centimeters and diagonal AC is 11 centimeters. Use the diagram below for questions 7-10.

7. The base of triangle ABC is 11 cm. and its corresponding height is 6 units. Therefore, the area of triangle ADC is 33.
 $\left(\frac{1}{2} \cdot 6 \cdot 11\right)$

8. The base of triangle ADC is 11 cm. and its corresponding height is 6 units. Therefore, the area of triangle ADC is 33.
 $\left(\frac{1}{2} \cdot 6 \cdot 11\right)$

9. The area of the rhombus is $33 + 33 = 66$ cm^2 .



10. Use the formula for area of a rhombus to verify your solution is correct.

$$\text{Area} = \frac{1}{2}(d_1)(d_2).$$

$$\text{Area} = \frac{1}{2}(\underline{12})(\underline{11})$$

$$\text{Area} = \underline{66} \text{ units}^2$$

Grade 6 Accelerated
Unit 7
Lesson 5 Homework

Name Answer Key
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1. The formula to find the area of a parallelogram is

$$A = b \cdot h$$

2. The formula to find the area of a trapezoid is

$$A = \frac{1}{2} \cdot h \cdot (b_1 + b_2)$$

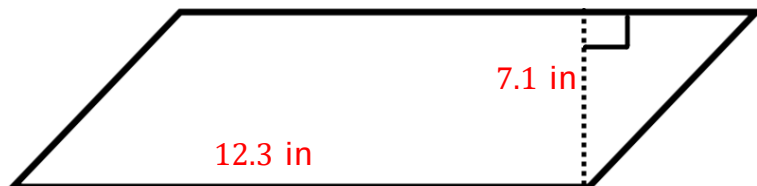
3. The formula to find the area of a rhombus is

$$A = \frac{1}{2} \cdot d_1 \cdot d_2$$

For questions 4-6, label each figure with the given dimensions, then calculate the area using the appropriate formula.

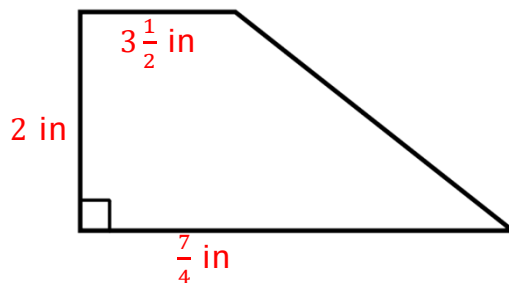
4. $b = 12.3$ in
 $h = 7.1$ in
 $12.3 \cdot 7.1$

$$\text{Area} = 87.33 \text{ in}^2$$



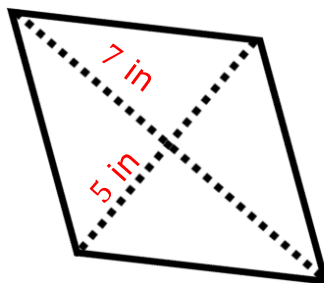
5. $b_1 = 3\frac{1}{2}$ in
 $b_2 = \frac{7}{4}$ in
 $h = 2$ in
 $\frac{1}{2} \cdot 2 \cdot \left(3\frac{1}{2} + \frac{7}{4}\right)$

$$\text{Area} = 5\frac{1}{4} \text{ in}^2$$



6. $d_1 = 7$ in
 $d_2 = 5$ in
 $\frac{1}{2} \cdot 7 \cdot 5$

$$\text{Area} = 17.5 \text{ in}^2$$



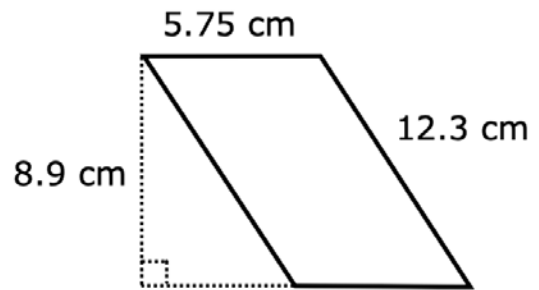
7. Zoey made an error in her work. Identify the error and then find the area of the parallelogram.

$$\text{Area} = (5.75)(12.3)$$

$$\text{Area} = 70.725 \text{ cm}^2$$

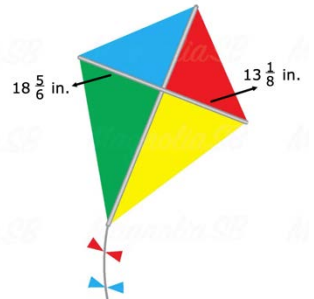
Zoey did not use the actual height; she used the slant height.

$$A = 5.75(8.9) = 51.175 \text{ cm}^2$$

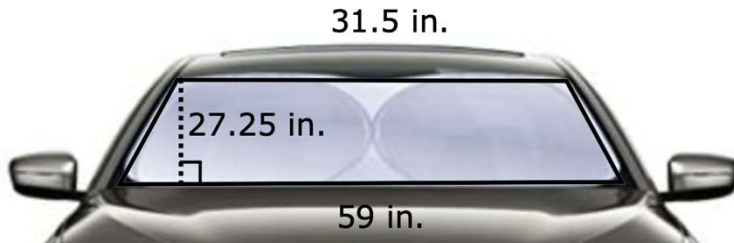


8. Kai and his father are constructing a kite to fly at the beach that is in the shape of a rhombus. The spars on the kite measure $13\frac{1}{8}$ inches and $18\frac{5}{6}$ inches. How many square inches (in^2) of fabric will be needed to construct the kite?

$$A = \frac{1}{2} \left(13\frac{1}{8} \right) \left(18\frac{5}{6} \right) = 123\frac{19}{32} \text{ in}^2$$



9. The windshield of a car is in the shape of a trapezoid. Find the area of the glass used to make the windshield.



$$A = \frac{1}{2} \cdot 27.25 \cdot (31.5 + 59)$$

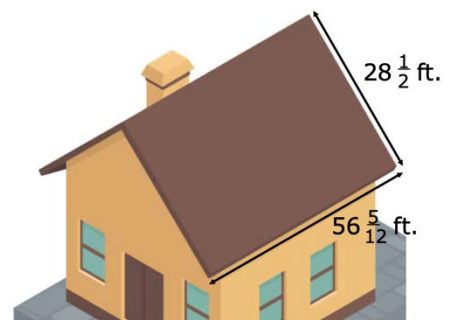
$$= \frac{1}{2} \cdot 27.25 \cdot 90.5$$

$$= 1233.0625 \text{ in}^2$$

10. Marc needs a new roof on his house and needs to measure how many roofing tiles are needed to get an accurate estimate from the roofing company. The roof has two sides. Each side is in the shape of a parallelogram with a base of $56\frac{5}{12}$ feet and a height of $28\frac{1}{2}$ feet. Find the how many square feet of roofing material is needed.

$$\text{Area} = 56\frac{5}{12} \left(28\frac{1}{2} \right) = 1607\frac{7}{8} \text{ or } 1607.875$$

$$\text{Total area} = 2(1607.875) = 3215.75 \text{ or } 3215\frac{3}{4} \text{ ft}^2$$



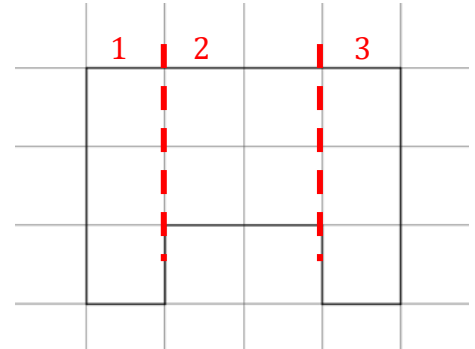
Grade 6 Accelerated
Unit 7
Lesson 6 Homework

Name Answer Key
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A composite figure is shown for questions 1–3. The length of each square is 1 inch (in.).

1. Draw lines to decompose the figure into rectangles.

Answers may vary.



2. Find the area of each rectangle. Include units.

$$\text{Area}_1 = 3 \cdot 1 = 3 \text{ in}^2$$

$$\text{Area}_2 = 2 \cdot 2 = 4 \text{ in}^2$$

$$\text{Area}_3 = 3 \cdot 1 = 3 \text{ in}^2$$

3. Find the sum of the areas of the rectangles to find the area of the composite figure. Include units.

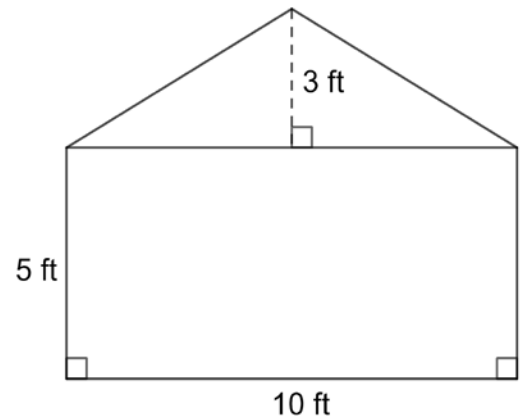
$$3 + 4 + 3 = 10 \text{ in}^2$$

A composite figure measured in feet (ft) is shown for questions 4–5.

4. Find the area of the triangle and the rectangle. Include units.

$$\text{Triangle} \rightarrow 0.5 \cdot 10 \cdot 3 = 15 \text{ ft}^2$$

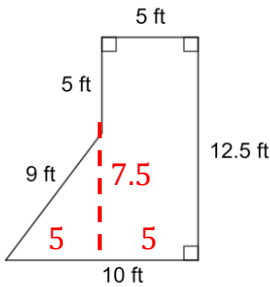
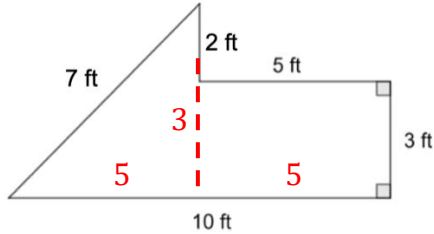
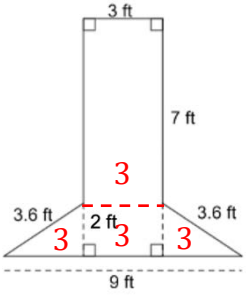
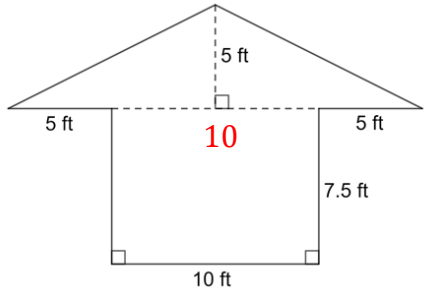
$$\text{Rectangle} \rightarrow 5 \cdot 10 = 50 \text{ ft}^2$$



5. Find the sum of the areas of the rectangle and triangle to find the area of the composite figure. Include units.

$$15 + 50 = 65 \text{ ft}^2$$

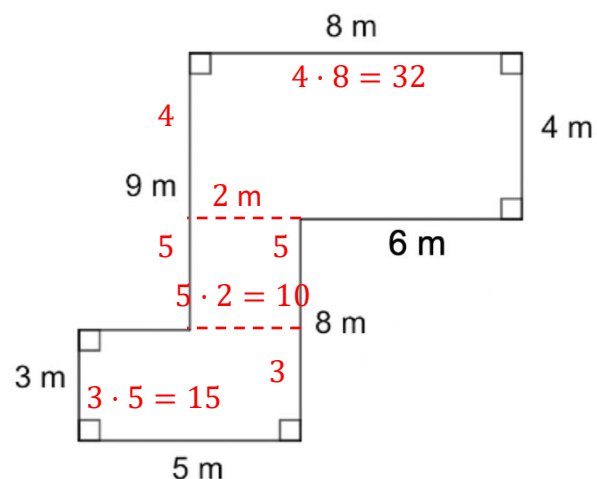
For each composite figure, find the area in square feet.

6. 	7. 
Triangle $\rightarrow 5 \cdot 7.5 \cdot 0.5 = 18.75$ Rectangle $\rightarrow 5 \cdot 12.5 = 62.5$ Total $\rightarrow 18.75 + 62.5 = 81.25$ Area = <u>81.25 ft²</u>	Triangle $\rightarrow \frac{1}{2} \cdot 5 \cdot 5 = 12.5 \text{ ft}^2$ Rectangle $\rightarrow 5 \cdot 3 = 15 \text{ ft}^2$ $12.5 + 15 = 27.5$ Area = <u>27.5 ft²</u>
8. 	9. 
Triangle₁ $\rightarrow 3 \cdot 2 \cdot \frac{1}{2} = 3$ $3 + 27 + 3 = 33$ Rectangle $\rightarrow 3 \cdot 9 = 27$ Triangle₂ $\rightarrow 3 \cdot 2 \cdot \frac{1}{2} = 3$ Area = <u>33 ft²</u>	Triangle $\rightarrow \frac{1}{2} \cdot 20 \cdot 5 = 50$ Rectangle $\rightarrow 10 \cdot 7.5 = 75$ $75 + 50 = 125$ Area = <u>125 ft²</u>

10. The local university is planting a garden. A drawing of the garden is shown. What is the area of the garden in square meters?

Rectangle₁ $\rightarrow 4 \cdot 8 = 32 \text{ m}^2$
 Rectangle₂ $\rightarrow 2 \cdot 5 = 10 \text{ m}^2$
 Rectangle₃ $\rightarrow 3 \cdot 5 = 15 \text{ m}^2$

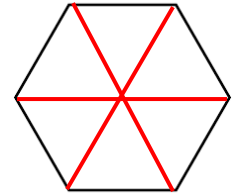
$32 + 10 + 15 = 57 \text{ m}^2$



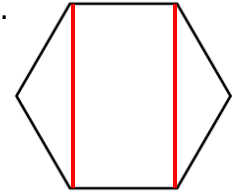
Grade 6 Accelerated
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Lesson 7 Homework

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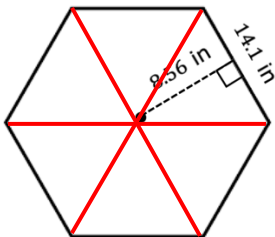
1. Decompose the regular hexagon into only triangles.



2. Decompose the regular hexagon into triangles and quadrilaterals.
Answers may vary.

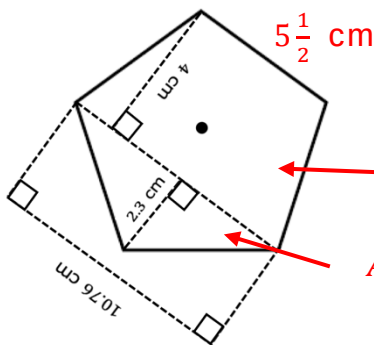


3. A hexagon is given in inches (in). Each side of the hexagon has a length of 14.1 inches and a perpendicular distance from the center to the midpoint of a side to be 8.56 inches. Find the area of the regular hexagon rounded to the nearest square inch.



$$A = 6 \left(\frac{1}{2} \cdot 8.56 \cdot 14.1 \right) = 6(60.348) \\ = 362 \text{ in}^2$$

4. Find the area of the regular pentagon that has a side length of $5\frac{1}{2}$ cm.



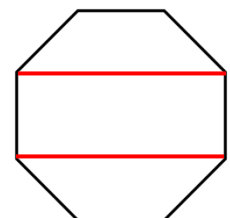
$$A = \frac{1}{2} \cdot 4 \cdot \left(5\frac{1}{2} + 10.76 \right) = 32.52 \text{ cm}^2$$

$$A = \frac{1}{2} \cdot 2.3 \cdot 10.76 = 12.374 \text{ cm}^2$$

$$\text{Total Area} = 32.52 + 12.374 \\ = 44.894 \text{ cm}^2$$

5. Decompose the regular octagon into three quadrilaterals.

Angle of lines may vary.



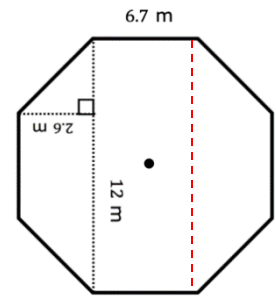
6. An octagon is shown with lengths in meters (m). What is the area of the octagon?

$$A_{trap} = \frac{1}{2} \cdot 2.6 \cdot (6.7 + 12) = 24.31$$

$$2 \text{ trapezoids} \rightarrow 2(24.31) = 48.62$$

$$A_{rect} = 6.7(12) = 80.4$$

$$\text{Total Area} = 48.62 + 80.4 = 129.02 \text{ m}^2$$



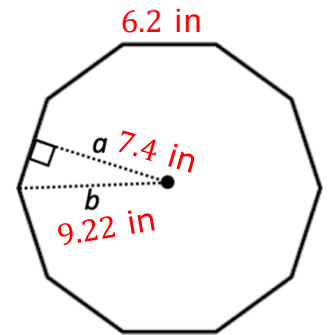
7. Label the regular decagon with the following dimensions.
Side length = 6.2 in.

$$a = 7.4 \text{ in.}$$

$$b = 9.22 \text{ in.}$$

Which of the dimensions is not needed to find the area of the decagon?

b

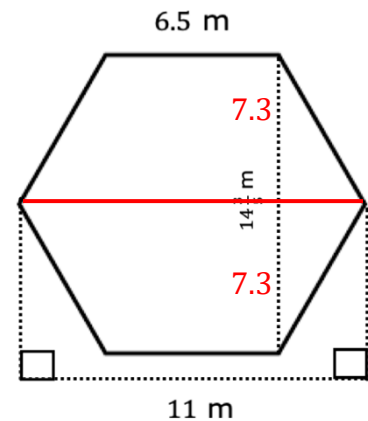


8. Find the area of the decagon from Question 7 in square inches.

$$A = 10 \left(\frac{1}{2} \cdot 7.4 \cdot 6.2 \right) = 229.4 \text{ in}^2$$

9. Myles found the area of the hexagon. His work is shown.

Myles' Work
Area of Trapezoid = $7.3(11 + 6.5)$ $A = 127.75 \text{ m}^2$
Area of hexagon = $2(127.75)$ $A = 255.5 \text{ m}^2$



Write a note to Myles explaining the error he made.

The area of a trapezoid is found by taking half of the height multiplied by the sum of the bases. He forgot to multiply by $\frac{1}{2}$.

10. Find the area of the hexagon in square meters.

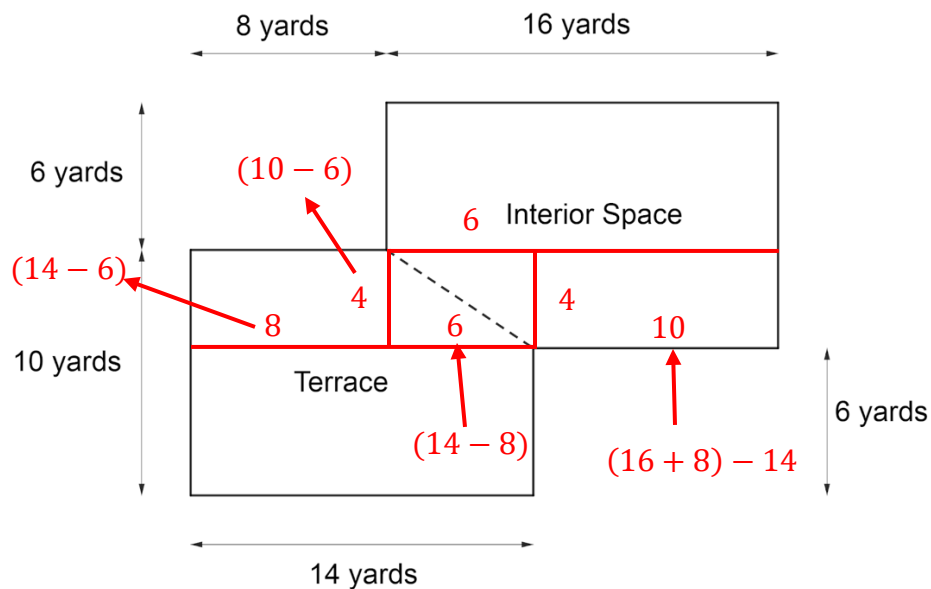
$$A_{trap} = \frac{1}{2} \cdot 7.3(11 + 6.5) = 63.875$$

$$\text{Total area} = 2(63.875) = 127.75 \text{ m}^2$$

Grade 6 Accelerated
Unit 7
Lesson 8 Homework

Name Answer Key
Date _____
Period _____

A local restaurant consists of both interior space and an outdoor terrace.



- Decompose the floorplan into triangles and quadrilaterals.
Answers may vary

- Find the area of the terrace.

$$(14 \cdot 6) + (8 \cdot 4) \rightarrow \left(\frac{1}{2} \cdot 6 \cdot 4\right)$$

$$84 + 32 + 12 = 128 \text{ yd}^2$$

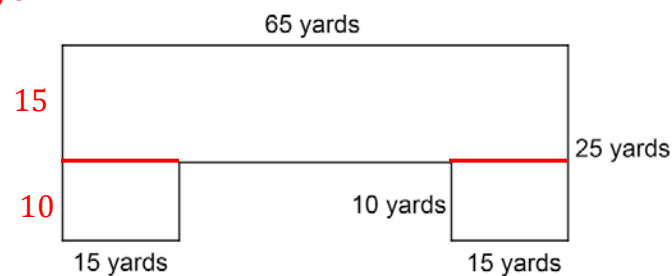
- Find the area of the interior space.

$$(6 \cdot 16) + (10 \cdot 4) + \left(\frac{1}{2} \cdot 6 \cdot 4\right)$$

$$96 + 40 + 12 = 148 \text{ yd}^2$$

- Find the area of the entire restaurant.

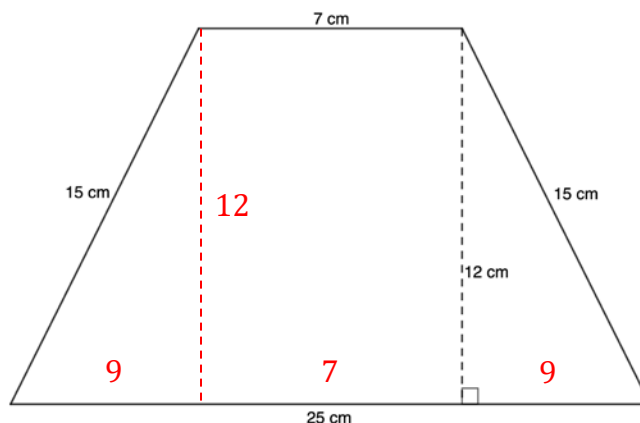
$$128 + 148 = 276 \text{ yd}^2$$



5. Decompose the image into quadrilaterals.

6. Find the area of the composite figure.

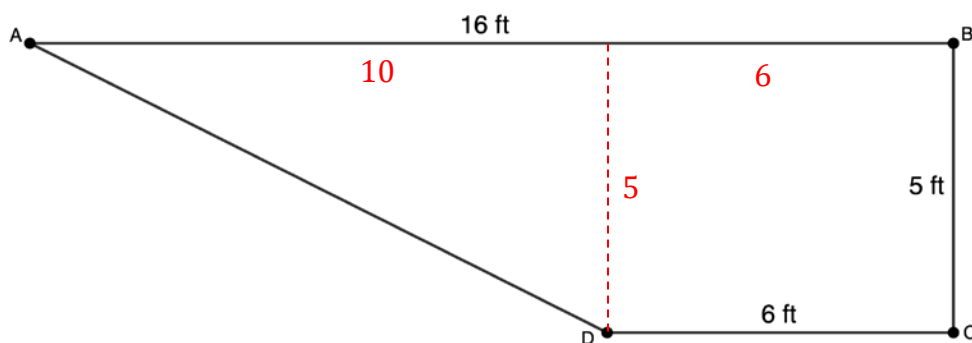
$$(10 \cdot 15) + (10 \cdot 15) + (15 \cdot 65) = 150 + 150 + 975 = 1275 \text{ yd}^2$$



$$25 - 7 = \frac{18}{2} = 9$$

7. Decompose the image into triangles and quadrilaterals.

8. Find the area of the image. $2 \left(\frac{1}{2} \cdot 9 \cdot 12 \right) + (7 \cdot 12) = 192 \text{ cm}^2$



9. Decompose the image into triangles and quadrilaterals.

10. Find the area of the composite figure.

$$\left(\frac{1}{2} \cdot 10 \cdot 5 \right) + (6 \cdot 5) = 25 + 30 = 55 \text{ ft}^2$$